

Problem Set 8 for PDEs. Due Friday Nov. 28 at 11:59 PM.

Problem 0: Please hand-write the numbers 0 through 9, scan, and compress each down to a 28x28 array of pixel brightnesses that go from 0 to 255. Put into a single array of size (10,28,28) and save as 'digits_[your student id]-week8.npy'. You can get a half bonus point if you write out 5 sets and save as a (50,28,28) array with the numbers ordered (0..9,0..9,0..9), and a full bonus point if you write out ten sets.

Problem 1: Earlier in the course, I claimed that the leapfrog scheme preserves energy. Show that this is true as long as the CFL condition is satisfied. Recall that the leapfrog scheme is:

$$\frac{f(t+dt, x) - f(t-dt, x)}{2dt} = -v \frac{f(t, x+dx) - f(t, x-dx)}{2dx}$$

In particular, we calculate the spatial derivative at a point halfway between the times at which we evaluate the solution. You may assume the solution will look something like $f(x, t) = \xi^t \exp(ikx)$ where in general ξ will be complex and a function of k . You may wish to consult Numerical Recipes.

Problem 2: In class, we wrote down that the Green's function for Poisson's equation was $1/r$ (in 3D) or $\ln(r)$ (in 2D). In reality, this is close but not quite right because of the effects of discretization. What we actually want is the function that has $\nabla^2 V = 0$ except at the origin. In class we specified the boundary conditions at the edge of our region, but for the Green's function, we don't want that. We'd rather have the Green's function for a point charge isolated in an infinite space. Since we can't do an infinite box, we'll instead pick a finite box. We then set the boundary conditions at the edge of the box so that the charge one cell away from the edge is zero, with the understanding that we don't use the actual edge of our box.

Using either relaxation or conjugate-gradient, solve for the actual Green's function for Poisson's equation in 2D. Normalize such that both V and ρ at the origin are 1 (where ρ is defined to be my potential minus the average of my neighbors). You should be able to convince yourself that V one cell away from the charge is zero, and if all has gone well, the potential five cells away from the charge should be around -1.05. What is its value exactly? Please also plot the inferred charge (not including the outer edge) to show your solution indeed has zero charge everywhere except the center. Please do this for a box size of at least 257.

Notes: I found it easier to put the charge at $(n/2, n/2)$ instead of $(0, 0)$. I also used an odd box size since then the edges should all be the same.

Finally, if you hit strange behavior, think carefully about the corners. If you put the charge at $(n/2, n/2)$, you should have a special condition for $V[1, 0]$ etc. that is different from $V[2 : -2, 0]$ etc.

It might be painful to generate the actual Green's function, but once we have it, we have it for all time.

Problem 3: Let's use conjugate gradient to solve Laplace's equation with boundary conditions using our new Green's functions. Since we can write a boundary-free Green's function, we can remove the outer shell of our box to find the potential from something in free space. We'll stick to two dimensions and use a square conductor, as we used for the in-class coax example. As a reminder, the square conductor is $1/4$ the width of the box, centered on the box center.

part a: Now that we know what the potential from a point charge is, we can calculate the potential everywhere in space from an arbitrary charge distribution by convolving the charge by our hard-won but eternal Green's function. We can write this as $V = \mathcal{G} \circledast \rho$, where $\mathcal{G}$ is the Green's function. Even though we don't usually think of it that way, as we saw in class this is indeed a matrix equation and so can be solved using conjugate-gradient (or any other tool you care to use). In this case, though, we start with the potential on some surfaces, and want to find the charge distribution on those same surfaces. Write a conjugate-gradient solver that solves for ρ on a mask given V on that mask. Use your solver to find the charge on a square box held at a potential of 1. Plot the charge density along one side of the box.

part b: Now that you have the charge, show the potential everywhere in space. How close to constant is the potential in the interior of the box? Now plot the x - and y -components of the field just outside the box. Do they agree with what you expect? As a reminder, the boundary conditions are that the field is perpendicular to any equipotential, and that standard lore says that fields are stronger near corners.

Final Comments: The easiest way to solve this is to find the potential everywhere in space and then read it out along the mask. For relatively small masks, though, there's no guarantee that this is the fastest way - a brute force summation may win in some cases. Also note that we got away with not having to specify an outer boundary condition. As long as you pulled a large enough region that the potential from your boundaries doesn't wrap around onto itself (if you're using an FFT) then there's no edge where boundaries are specified. This is generally a good thing, since our usual state of affairs is not to be sitting near the center of a grounded,

conducting box. Finally, you may have noticed that I played a bit fast and loose with zero points. In 3D, it's perfectly sensible to set the potential at infinity to zero, but we can't do that in 2D since $\ln(x)$ unhappily diverges at both large and small x . In some sense then "hold a box at fixed potential" doesn't even make sense (fixed relative to what?). However, by setting the potential of our Green's function to 1 at the center, we have implicitly set a zero point.